Reusing classes

Agenda

1. Benefits of reusing existing entities
2. Has-a relationship and its types
3. Containment
4. Association
5. Comparing containment with association

1. Benefits of reusing existing entities

1. Reusing existing class for creating new classes
2. Increases productivity of software development
3. Increases reliability as pre-tested classes are re-used

2. Has-a relationship and its types

1. Examples of has-a relationship in real life:
    - Car has engine
    - Book has pages
    - Mobile has SIM
2. Types of has-a relationship
    - Physical binding (Tight coupling)
    - Logical binding (Loose coupling)

3. Containment

1. One object physically connected to other object
2. Component created with container and destroyed with container
3. Changes in component directly affect container
4. Not recommended as far as possible
5. Examples
   - Car and Engine
   - Mobile and SIM

3. Containment

1. Steps to implement containment
    - Create object of component as data member of container
    - Accepts all data for component in the constructor of container.
    - Distribute this data to components using MIL
    - Reuse functionality of component inside container

- Order of execution of constructor when container is created
- Order of execution of destructor when container is destroyed

4. Association

1. One object NOT physically connected to other object. It is just associated logically
2. Component may exist even before the container and can still exist after the container is destroyed
3. Changes in component don't directly affect container
4. Recommended as far as possible
5. Examples
   - Car and Driver
   - Mobile and Charger
   - Person and Aadhar card

4. Association

1. Steps to implement association
    - Create pointer (or reference) of component as data member of container
    - Accepts all data for component in the constructor of container.
    - Distribute this data to components while creating it dynamically
    - Reuse functionality of component inside container using pointer to component

4. Comparing containment with association

1. Physically connect v/s logically connected
2. Dependent v/s Independent life span
3. Rigid v/s flexible